

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	2 July 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	4 Marks

Data Layer

Component	Description
Dataset	food_coded.csv – structured CSV dataset capturing dietary and health info
Storage Format	Flat file (CSV) loaded locally for analysis
Tools Used	Excel (for cleaning & preprocessing if needed)

Data Processing Layer

Tool/Technology	Purpose
Excel	Initial cleaning or quick field review before importing to Tableau

Visualization & Analytics Layer

Tool/Technology	Purpose
Tableau	Main tool for interactive data visualization and dashboard creation
Tableau Features	KPI cards, Bar Charts, Pie Charts, Heatmaps, Highlight Tables, Forecasts
Calculated Fields	Used for: Consumption, Food Selection

User Interaction Layer

Feature	Role
Interactive Dashboards	Users can filter data by gender, risk, and exercise levels
Parameter Controls	Customize target values
Alerts & KPIs	Instant insight into nutrition deficiencies and trends

User Roles	Admin, Nutritionist, Cafeteria Staff – each interacts with filtered views
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Security & Sharing

Feature	Notes
Tableau Public	Public dashboards (for non-sensitive data)
Tableau Server	Optional upgrade for secured, role-based access
Export Options	PDF reports, public link sharing, dashboard embedding

Justification for Technology Choices

- **Tableau** is chosen for its ease of use, powerful visualization capabilities, and ability to integrate dynamic, calculated insights without extensive coding.
- **CSV Format** keeps the project simple and lightweight for academic or prototype environments.
- **Excel/Python** enables initial analysis and advanced preprocessing before visual design begins.